# DFlash on TPU

Speculative Decoding Benchmark Report
Qwen3-4B + DFlash-b16 | TPU v4 vs v5p

## Overview

| PROPERTY | VALUE |
|---|---|
| Target Model | `Qwen/Qwen3-4B` (3.4B params, 36 layers) |
| Draft Model | `z-lab/Qwen3-4B-DFlash-b16` (diffusion drafter) |
| Block Size | 16 (draft 15 tokens per step) |
| Benchmarks | 9 datasets across math, code, chat |
| Samples | 8 per dataset, 256 max tokens, temp=0.0, 1 warmup |
| Runtime | Docker `vllm/vllm-tpu:latest` (JAX 0.8.1, Flax 0.11.1) |

> **DFlash** uses a diffusion-style draft model that proposes token blocks in O(1) forward passes, achieving high acceptance rates on structured tasks (math, code).

# ## V5P Results — All 9 Datasets

| DATASET | CAT | BASELINE | DFLASH | SPEEDUP | TAU | MATCH |
|---|---|---|---|---|---|---|
| gsm8k | math | 6.35 ms | 2.12 ms | 3.00x | 5.40 | 5/8 |
| math500 | math | 6.37 ms | 1.29 ms | 4.93x | 8.80 | 3/8 |
| aime24 | math | 6.31 ms | 1.77 ms | 3.57x | 6.48 | 2/8 |
| aime25 | math | 6.38 ms | 1.88 ms | 3.39x | 6.14 | 3/8 |
| humaneval | code | 6.31 ms | 1.99 ms | 3.17x | 5.76 | 4/8 |
| mbpp | code | 6.40 ms | 1.96 ms | 3.27x | 6.16 | 3/8 |
| mt-bench | chat | 6.27 ms | 2.74 ms | 2.29x | 3.87 | 1/8 |
| alpaca | chat | 6.29 ms | 3.86 ms | 1.63x | 2.86 | 4/8 |
| swe-bench | code | 6.32 ms | 3.40 ms | 1.86x | 3.35 | 0/8 |

**Match** = exact token match between DFlash and autoregressive baseline. Mismatches are bf16 floating-point divergence (batch-16 verify vs single-token), not correctness loss.

## V5P Results — By Category

| CATEGORY | AVG SPEEDUP | AVG TAU | AVG DFLASH TPS | BEST DATASET |
|---|---|---|---|---|
| Math (4) | 3.72x | 6.71 | 585.4 | math500 (4.93x) |
| Code (3) | 2.77x | 5.09 | 435.7 | mbpp (3.27x) |
| Chat (2) | 1.96x | 3.37 | 312.2 | mt-bench (2.29x) |
| All (9) | 3.02x | 5.42 | 474.8 | |

**Why math >> chat?** Structured math reasoning has highly predictable token sequences (equations, step numbering). Chat has more creative variation, reducing draft acceptance rate.

# ## V5P Results — Throughput (TPS)

| DATASET | BASELINE TPS | DFLASH TPS | GAIN |
|---|---|---|---|
| gsm8k | 157.5 | 471.9 | +200% |
| math500 | 156.9 | 772.7 | +392% |
| aime24 | 158.4 | 565.0 | +257% |
| aime25 | 156.9 | 532.0 | +239% |
| humaneval | 158.4 | 501.9 | +217% |
| mbpp | 156.4 | 510.7 | +226% |
| mt-bench | 159.4 | 365.2 | +129% |
| alpaca | 158.9 | 259.2 | +63% |
| swe-bench | 158.2 | 294.5 | +86% |

**Peak throughput**: **772.7 TPS** on math500 — the draft model achieves tau=8.80 (nearly 9 tokens accepted per draft block of 15).

# ## V5P vs V4

Hardware generation comparison on identical benchmarks

# ## V5P vs V4 — Latency (TPOT)

| DATASET | V5P BASELINE | V4 BASELINE | V5P DFLASH | V4 DFLASH |
|---|---|---|---|---|
| gsm8k | 6.35 ms | 10.75 ms | 2.12 ms | 3.20 ms |
| math500 | 6.37 ms | 10.73 ms | 1.29 ms | 1.88 ms |
| aime24 | 6.31 ms | 10.68 ms | 1.77 ms | 2.72 ms |
| aime25 | 6.38 ms | 10.74 ms | 1.88 ms | 2.57 ms |
| humaneval | 6.31 ms | 10.81 ms | 1.99 ms | 3.01 ms |
| mbpp | 6.40 ms | 10.66 ms | 1.96 ms | 3.27 ms |
| mt-bench | 6.27 ms | 10.69 ms | 2.74 ms | 4.15 ms |
| alpaca | 6.29 ms | 10.65 ms | 3.86 ms | 5.47 ms |
| swe-bench | 6.32 ms | 10.65 ms | 3.40 ms | 8.42 ms |

V5P baseline is **1.69x** faster than V4 across the board. V5P DFlash is **1.55x** faster in absolute latency.

# ## V5P vs V4 — Speedup & Tau

| DATASET | V5P SPEEDUP | V4 SPEEDUP | V5P TAU | V4 TAU |
|---|---|---|---|---|
| gsm8k | 3.00x | 3.36x | 5.40 | 5.17 |
| math500 | 4.93x | 5.70x | 8.80 | 8.72 |
| aime24 | 3.57x | 3.93x | 6.48 | 6.32 |
| aime25 | 3.39x | 4.18x | 6.14 | 6.48 |
| humaneval | 3.17x | 3.59x | 5.76 | 5.75 |
| mbpp | 3.27x | 3.26x | 6.16 | 5.95 |
| mt-bench | 2.29x | 2.58x | 3.87 | 3.84 |
| alpaca | 1.63x | 1.95x | 2.86 | 2.83 |
| swe-bench | 1.86x | 1.27x | 3.35 | 3.28 |

**Speedup ratios are lower on V5P** because the baseline improved proportionally more than DFlash. However, **absolute DFlash performance is better** on V5P in every dataset. Tau is hardware-independent (model quality metric).

## V5P vs V4 — Summary

| METRIC | V5P | V4 | DELTA |
|---|---|---|---|
| Avg Baseline TPOT | 6.33 ms | 10.71 ms | 1.69x faster |
| Avg DFlash TPOT | 2.44 ms | 3.85 ms | 1.58x faster |
| Avg Speedup | 3.02x | 3.36x | -10% ratio |
| Avg Tau | 5.42 | 5.37 | +0.9% |
| Best DFlash TPOT | 1.29 ms | 1.88 ms | 1.46x faster |
| Peak TPS | 772.7 | 531.5 | +45% |

**Key insight**: V5P delivers higher absolute throughput but lower speedup *ratios* because autoregressive decoding also benefits from faster hardware. DFlash's value increases with model size (where baseline is slower).

## V5P vs GPU Paper

DFlash paper (arXiv 2025) reported GPU results on A100

# ## V5P vs GPU Paper — Math Benchmarks

| DATASET | V5P TAU | GPU TAU | TAU % | V5P SPEEDUP | GPU SPEEDUP |
|---|---|---|---|---|---|
| gsm8k | 5.40 | 6.53 | 82.7% | 3.00x | 5.15x |
| math500 | 8.80 | 7.84 | 112.2% | 4.93x | 6.09x |
| aime24 | 6.48 | 7.27 | 89.1% | 3.57x | 5.68x |
| aime25 | 6.14 | 6.64 | 92.5% | 3.39x | 5.21x |
| Average | 6.71 | 7.07 | 94.9% | 3.72x | 5.53x |

V5P TPU achieves **94.9% of GPU paper tau** on math benchmarks — nearly full acceptance rate parity. Speedup gap is due to TPU's already-fast baseline (6.3ms vs GPU's slower autoregressive path).

Math500 **exceeds** GPU tau (112.2%), likely due to sampling variance at 8 samples.

# ## Output Quality

| DATASET | EXACT MATCH | RATE | ANSWER MATCH | NOTES |
|---|---|---|---|---|
| gsm8k | 5/8 | 62.5% | 2/3 | High match on structured math |
| math500 | 3/8 | 37.5% | 2/2 | Answers always correct |
| aime24 | 2/8 | 25.0% | N/A | Competition problems diverge early |
| aime25 | 3/8 | 37.5% | N/A | Similar to aime24 |
| humaneval | 4/8 | 50.0% | N/A | Code structure preserved |
| mbpp | 3/8 | 37.5% | N/A | Variable names may differ |
| mt-bench | 1/8 | 12.5% | N/A | Creative text diverges easily |
| alpaca | 4/8 | 50.0% | N/A | Short responses match better |
| swe-bench | 0/8 | 0.0% | N/A | Long context, early divergence |

**Exact match** = DFlash produces identical tokens to baseline. Mismatches come from bf16 batch-16 verify vs single-token decode producing different floating-point accumulation. This flips argmax at decision boundaries — **not a correctness issue**.

## Environment & Reproducibility

# ## Hardware & Software

| COMPONENT | V5P | V4 |
|---|---|---|
| TPU Type | v5p-8 (4 chips) | v4-8 (4 chips) |
| Topology | 2×2×1×1 | 2×2×1 |
| Device Interface | `/dev/vfio/[0-3]` | `/dev/accel/[0-3]` |
| `/dev/shm` | 221 GB | 330 GB |
| Python | 3.12 (Docker) | 3.12 (Docker) |
| JAX | 0.8.1 | 0.8.1 |
| Flax | 0.11.1 | 0.11.1 |
| Docker Image | `vllm/vllm-tpu:latest` | `vllm/vllm-tpu:latest` |

Both V4 and V5P benchmarks use the same Docker image and identical code ( `benchmarks/standalone_dflash.py` ). Results are directly comparable.

## Reproduce

```
# Activate environment
source ~/venv/bin/activate
export PYTHONPATH="$HOME/tpu-spec-decode/vllm:$HOME/tpu-spec-decode/tpu-inference"

# Run a single benchmark (Docker)
sudo docker run --rm --privileged --network host --ipc host \
  --tmpfs "/tmp:rw,size=80g" \
  -e HF_HOME=/hf-cache -e HUGGINGFACE_HUB_CACHE=/hf-cache/hub \
  -e PYTHONPATH=/workspace/tpu-spec-decode/vllm:/workspace/tpu-spec-decode/tpu-inference \
  -v /dev/shm/hf-cache:/hf-cache \
  -v /home/$USER/tpu-spec-decode:/workspace/tpu-spec-decode \
  -w /workspace/tpu-spec-decode \
  vllm/vllm-tpu:latest \
  python3 benchmarks/standalone_dflash.py \
    --target-model Qwen/Qwen3-4B \
    --draft-model z-lab/Qwen3-4B-DFlash-b16 \
    --dataset gsm8k --max-samples 8 --max-new-tokens 256 \
    --temperature 0.0 --warmup 1 \
    --output-json /output/result.json
```

# ## File Layout

```
results/
  v4/                              # V4 benchmark data (28 files)
    standalone_*.json              # Raw results per dataset
    standalone_all_benchmarks.csv  # Summary table
    standalone_vs_gpu_paper.csv    # V4 vs GPU paper
    report.md                      # V4 report
  v5p/                             # V5P benchmark data (12 files)
    standalone_*.json              # Raw results per dataset
    standalone_all_benchmarks.csv  # Summary table (with accuracy)
    standalone_vs_v4.csv           # V5P vs V4 comparison
    standalone_vs_gpu_paper.csv    # V5P vs GPU paper
  report.md                        # This report (Marp slides)
```

> Raw JSON files contain per-sample data: token outputs, acceptance histograms, per-position acceptance rates, quality comparisons.

# ## PR Validation (pr/dflash)

Upstream PR benchmarked on TPU v5p vs GPU paper (A100)

# ## PR Validation — TPU v5p vs GPU (A100)

| DATASET | CAT | TPU TAU | GPU TAU | TAU % | TPU SPEEDUP | GPU SPEEDUP |
|---|---|---|---|---|---|---|
| gsm8k | math | 5.40 | 6.53 | 82.7% | 2.95x | 5.15x |
| math500 | math | 8.80 | 7.84 | 112.2% | 4.90x | 6.09x |
| aime24 | math | 6.48 | 7.27 | 89.1% | 3.49x | 5.68x |
| aime25 | math | 6.14 | 6.64 | 92.5% | 3.32x | 5.21x |
| humaneval | code | 5.76 | — | — | 3.09x | — |
| mbpp | code | 6.16 | — | — | 3.41x | — |
| mt-bench | chat | 3.87 | — | — | 2.38x | — |
| alpaca | chat | 2.86 | — | — | 1.65x | — |
| swe-bench | code | 3.35 | — | — | 1.97x | — |

Results from `pr/dflash` branch (aaronzhfeng/tpu-inference). Math tau is **94.9%** of GPU paper on TPU — near-parity in draft quality. Speedup gap is architectural: TPU's faster autoregressive baseline compresses the ratio.

# ## PR Validation — Summary

| METRIC | TPU V5P (PR) | GPU A100 (PAPER) |
|---|---|---|
| Math Avg Tau | 6.71 | 7.07 |
| Math Avg Speedup | 3.67x | 5.53x |
| Math Tau Parity | 94.9% | 100% |
| Peak Tau | 8.80 (math500) | 7.84 (math500) |
| Datasets Tested | 9 | 4 |
| Hardware | v5p-8, 4 chips | A100 80GB |

**DFlash on TPU matches GPU draft quality** — the acceptance rate gap is <6% on math. The speedup ratio difference is explained by TPU's 6.3ms autoregressive TPOT vs GPU's slower baseline. At K=16, DFlash already delivers **3.02x** average speedup. The real opportunity is scaling to K=64/128 where TPU verification cost stays flat.

# Summary

**3.02x** avg speedup on V5P across 9 benchmarks
**4.93x** peak on math500 | **772.7 TPS** peak throughput
**94.9%** of GPU paper tau on math benchmarks
**1.69x** faster baseline than V4 | **1.55x** faster DFlash